

Name: _____
Date: _____ Per: _____

Algebra 2
Unit 7: Simplifying Polynomials

Quiz 7-1: Polynomial Operations & Radicals

Multiple Choice — Version 2

Circle the letter of the best answer for each question.

For questions 1–4, use the polynomial: $5x^3 - 9 + 2x^5 - x$

1. What is the polynomial written in standard form?

A) $2x^5 + 5x^3 - x - 9$

C) $5x^3 - 9 + 2x^5 - x$

B) $-9 - x + 5x^3 + 2x^5$

D) $2x^5 - x + 5x^3 - 9$

2. What is the degree of the polynomial?

A) 3

B) 4

C) 5

D) 9

3. How is the polynomial classified by number of terms?

A) Monomial

B) Binomial

C) Trinomial

D) Polynomial (4+ terms)

4. How is the polynomial classified by degree?

A) Quadratic

B) Cubic

C) Quartic

D) Quintic

For questions 5–7, use the polynomial: $8 - 4n^2 + 6n^4$

5. What is the polynomial written in standard form?

A) $8 - 4n^2 + 6n^4$

C) $-4n^2 + 6n^4 + 8$

B) $6n^4 - 4n^2 + 8$

D) $6n^4 + 8 - 4n^2$

6. What is the degree and leading coefficient of the polynomial?

A) Degree 4, leading coefficient 6

C) Degree 2, leading coefficient -4

B) Degree 4, leading coefficient -4

D) Degree 6, leading coefficient 4

7. How is the polynomial classified by number of terms?

A) Monomial

B) Binomial

C) Trinomial

D) Polynomial (4+ terms)

Simplify. All answers should be in standard form with only positive exponents.

8. Simplify: $(-3a^2b^3)^3 \cdot 4a^2b$

A) $108a^8b^{10}$

C) $-108a^8b^{10}$

B) $-108a^6b^{10}$

D) $-12a^8b^{10}$

9. Simplify: $\frac{6x^{-2}y^3 \cdot 4x^4y}{8x^5y^2}$

A) $3x^3y^2$

B) $\frac{3y^2}{x^3}$

C) $\frac{3}{x^3y^2}$

D) $\frac{3y}{x^3}$

10. Simplify: $(8x^2 - 3x + 5) + (4x - 2x^2 + 7)$

A) $10x^2 + x + 12$

C) $6x^2 - 7x + 12$

B) $6x^2 + 7x + 12$

D) $6x^2 + x + 12$

11. Simplify: $(5k^4 + 2k^2 - 3) - (4k^2 - 6k^3 + 8)$

A) $5k^4 + 6k^3 + 6k^2 - 11$

C) $5k^4 - 6k^3 + 6k^2 + 5$

B) $5k^4 + 6k^3 - 2k^2 - 11$

D) $5k^4 - 6k^3 - 2k^2 - 11$

12. Simplify: $(x - 3)(3x^2 + 2x - 5)$

A) $3x^3 + 7x^2 - 11x - 15$

C) $3x^3 - 7x^2 - 11x + 15$

B) $3x^3 - 7x^2 + 11x + 15$

D) $3x^3 - 7x^2 - 11x - 15$

13. Simplify: $\frac{18m^8n^4 - 24m^5n^2 + 6m^3n}{6m^3n}$

A) $3m^{11}n^5 - 4m^8n^3 + 1$

C) $3m^5n^3 - 4m^2n - 1$

B) $12m^5n^3 - 18m^2n + 1$

D) $3m^5n^3 - 4m^2n + 1$

14. A triangle has a base of $12x^3y^2$ and a height of $5x^4y$. What is the area of the triangle in simplified form?

(Hint: Area of triangle = $\frac{1}{2}bh$)

A) $60x^7y^3$

C) $30x^{12}y^2$

B) $30x^7y^3$

D) $60x^{12}y^3$

Rewriting and Simplifying Radicals

(TEKS 2A.7B, 2A.7G)

Convert between radical and rational exponent form, then simplify.

15. Which expression is equivalent to $\sqrt[4]{x}$?
- A) x^4 B) $x^{1/4}$ C) $4x$ D) $\frac{x}{4}$
16. Which expression is equivalent to $\sqrt[5]{m}$?
- A) m^5 B) $m^{1/5}$ C) $5m$ D) m^{-5}
17. Which expression is equivalent to $y^{3/4}$?
- A) $\sqrt[4]{y^3}$ B) $\sqrt[3]{y^4}$ C) $\frac{3y}{4}$ D) $\frac{3}{4}y$
18. What is the value of $16^{3/4}$?
- A) 4 B) 8 C) 12 D) 64
19. What is the value of $32^{2/5}$?
- A) 2 B) 4 C) $\frac{64}{5}$ D) 16
20. Simplify: $x^{1/3} \cdot x^{2/3}$
- A) $x^{2/9}$ B) x C) $x^{2/3}$ D) $2x^{1/3}$
21. Simplify: $a^{5/6} \cdot a^{1/6}$
- A) $a^{5/36}$ B) a C) $a^{1/2}$ D) $a^{6/6}$
22. Simplify: $(b^{1/3})^9$
- A) b^3 B) b^{27} C) $b^{1/27}$ D) $9b^{1/3}$
23. What is the value of $81^{-3/4}$?
- A) -27 B) 27 C) $\frac{1}{27}$ D) $-\frac{1}{27}$
24. Which expression is equivalent to $\sqrt[3]{x} \cdot \sqrt[6]{x}$?
- A) $x^{1/18}$ B) $x^{1/2}$ C) $x^{2/9}$ D) $x^{9/2}$
25. The expression $(8x^9)^{2/3}$ is equivalent to which of the following?

A) $2x^6$

B) $4x^6$

C) $4x^{27/2}$

D) $16x^6$

ANSWER KEY – TEACHER COPY

Version 2

Polynomial Operations (Questions 1–14)

- | | |
|------|-------|
| 1. A | 8. C |
| 2. C | 9. B |
| 3. D | 10. D |
| 4. D | 11. B |
| 5. B | 12. C |
| 6. A | 13. D |
| 7. C | 14. B |

Radicals & Rational Exponents (Questions 15–25)

- | | |
|-------|-------|
| 15. B | 21. B |
| 16. B | 22. A |
| 17. A | 23. C |
| 18. B | 24. B |
| 19. B | 25. B |
| 20. B | |

Detailed Solutions:

- $2x^5 + 5x^3 - x - 9$ (descending order of exponents)
- Degree 5 (highest exponent)
- 4 terms = polynomial
- Quintic (degree 5)
- $6n^4 - 4n^2 + 8$ (descending order)
- Degree 4, leading coefficient 6
- Trinomial (3 terms)
- $(-3a^2b^3)^3 \cdot 4a^2b = -27a^6b^9 \cdot 4a^2b = -108a^8b^{10}$
- $\frac{6x^{-2}y^3 \cdot 4x^4y}{8x^5y^2} = \frac{24x^2y^4}{8x^5y^2} = \frac{3y^2}{x^3}$

10. $(8x^2 - 3x + 5) + (4x - 2x^2 + 7) = 6x^2 + x + 12$
11. $(5k^4 + 2k^2 - 3) - (4k^2 - 6k^3 + 8) = 5k^4 + 6k^3 - 2k^2 - 11$
12. $(x - 3)(3x^2 + 2x - 5) = 3x^3 + 2x^2 - 5x - 9x^2 - 6x + 15 = 3x^3 - 7x^2 - 11x + 15$
13. $\frac{18m^8n^4 - 24m^5n^2 + 6m^3n}{6m^3n} = 3m^5n^3 - 4m^2n + 1$
14. Area = $\frac{1}{2}(12x^3y^2)(5x^4y) = \frac{1}{2}(60x^7y^3) = 30x^7y^3$
15. $\sqrt[4]{x} = x^{1/4}$ — The fourth root equals exponent of $\frac{1}{4}$
16. $\sqrt[5]{m} = m^{1/5}$ — The fifth root equals exponent of $\frac{1}{5}$
17. $y^{3/4} = \sqrt[4]{y^3}$ — Denominator is root, numerator is power
18. $16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8$
19. $32^{2/5} = (\sqrt[5]{32})^2 = 2^2 = 4$
20. $x^{1/3} \cdot x^{2/3} = x^{1/3+2/3} = x^{3/3} = x$ — Product Rule
21. $a^{5/6} \cdot a^{1/6} = a^{5/6+1/6} = a^{6/6} = a$ — Product Rule
22. $(b^{1/3})^9 = b^{(1/3) \cdot 9} = b^3$ — Power Rule
23. $81^{-3/4} = \frac{1}{81^{3/4}} = \frac{1}{(\sqrt[4]{81})^3} = \frac{1}{3^3} = \frac{1}{27}$
24. $\sqrt[3]{x} \cdot \sqrt[6]{x} = x^{1/3} \cdot x^{1/6} = x^{2/6+1/6} = x^{3/6} = x^{1/2}$
25. $(8x^9)^{2/3} = 8^{2/3} \cdot x^{9 \cdot 2/3} = 4 \cdot x^6 = 4x^6$